



Bangladesh University of Business and Technology (BUBT)
Faculty of Engineering & Applied Sciences (FEAS)
Department of Computer Science and Engineering (CSE)

Program: B.Sc. Engg. in Computer Science and Engineering

Session: Spring 2024

Level-Term: L1-T2

Course Code: CSE 111

Course Title: Object Oriented Programming Language

Credit Value: 3

Course Type: Core Course

Credit Hours: 3.00

Prerequisite: CSE 101

Total Marks: 100

COURSE TEACHER:

Name : Khondokar Oliullah

Classroom Code : vw2n6dt

Designation : Lecturer

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Specialization : Cloud Computing, Machine Learning, IoT

CLASS SCHEDULE:

Class Day	Class Hours	Room
Monday	11.45 am – 01.00 pm	B2/708
Wednesday	08:00 am – 09:15 pm	B2/810

COUNSELING HOURS:

Day	Counseling Hours	Room
Monday	09.15 am – 10.30 am	B3/502
Wednesday	09.15 am – 10.30 am	B3/502
Tuesday	1.30 pm – 02.45 pm	B3/502

Course Rationale	The Course- “Object Oriented Programming Language” provides in-depth coverage of object-oriented programming principles and techniques using C++. The course is designed to perform object-oriented programming to develop solutions to problems demonstrating the usage of control structures, modularity, I/O, and other standard language constructs. Also to demonstrate adeptness of object-oriented programming in developing solutions to problems demonstrating usage of data abstraction, encapsulation, and inheritance.
Course Objectives	It is expected that by the end of the course, the students will be able to prepare object-oriented design for small/medium scale problems, demonstrate the differences between traditional imperative design and object-oriented design, explain class structures as fundamental modular building blocks, understand the role of inheritance, polymorphism, dynamic binding and generic structures in building reusable code.
Course Outcomes (COs)	Upon completing this course students will be able to: CLO1: Describe Object Oriented Programming concepts and their features. CLO2: Understand how object-oriented programming concepts are used to solve a problem. CLO3: Apply OOP concepts (inheritance, interface, polymorphism, encapsulation, function overloading, operator overloading etc.) to solve different problems. CLO4: Analyze real life problem scenarios to design a solution using key features (abstraction, inheritance, polymorphism, encapsulation, exception handling, and template, multi-threading) of OOP using C++.

COs to POs Mapping	CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	√											
	CO2	√											
	CO3			√									
	CO4		√										
	CO	PO	Bloom's Domain / Level		Delivery Methods / Activities					Assessment Tools			
	CO1	PO1	Cognitive/Understanding		Class Lecture & Discussion					Midterm			
	CO2	PO1	Cognitive/Understanding		Class Lecture & Discussion					Midterm & Final			
	CO3	PO2	Cognitive/Applying		Class Lecture & Problem Solving					Midterm & Final			
	CO4	PO3	Cognitive/Analyzing		Class Lecture & Problem Solving					Final			

LECTURE PLAN (WEEKLY SCHEDULE)

Week	Selected Topics	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
1-2 (Week 1)	Introduction to Object-Oriented Programming, Features of OOP. Two versions of C++, C++ console I/O and Difference between C and C++, C++ comments, Classes a first look.	Lecture, Video Presentation	Class Test, Assignment, Written Examination	CO1
3-4 (Week 2)	Class and Objects: Constructor and Destructor Functions and Relation of Classes, Structures and Unions, Functions, Automatic inline functions and Constructor that take parameters, Object Pointers.	Lecture discussion with White Board and Multimedia, Problem Solving		CO1,CO2
5-6 (Week 3)	A closer look to classes, Assigning objects, Passing and Returning objects to and from a function and An introduction to Friend Function.			CO2,CO3
7-8 (Week 4)	Arrays of objects, Using Pointers to objects, The “This” pointer, Using New and Delete.			CO1, CO2
9-10 (Week 5)	Reference, Passing Reference of objects, Returning Reference, Independent reference and restrictions, Overloading Constructor Function and Creating and using a copy constructor.			CO2,CO3
11-12 (Week 6)	The overload Anachronism, Using Default Arguments, Overloading and ambiguity, Finding the address of an overloaded function			CO2
13-14 (Week 7)	Case Study & Review class for Semester Mid Examination.	Lectures, Questions and answers	Problem Solving & Quiz Test	CO2,CO3

Midterm Examination (29 Aug – 15 Sep 2024)

15-16 (Week 8)	Operator overloading: Overloading Binary operator, overloading the [] operator, Overloading unary operators, Overloading relational operators			CO2, CO3
17-18 (Week 9)	Friend operator functions and Overloading shorthand assignment operators, Overloaded assignment operator and copy constructor, Inheritance: Base Class access control			CO2, CO3

19-20 (Week 10)	Using protected members, Constructor, Destructor and inheritance, Multiple inheritance	Lecture discussion with White Board and Multimedia, Problem Solving	Class Test, Assignment, Problem Solving Ability and Written Examination	CO3,CO4
21-22 (Week 11)	Virtual Base Class, Pointers to derived classes, Introduction to Virtual Functions			CO2
23-24 (Week 12)	Pure virtual function, Abstract class, Applying run time polymorphism, Generic Function, Generic class			CO3, CO4
25-26 (Week 13)	Exception specification, Exception Handling			CO2, CO4
27-28 (Week 14)	ACM-ICPC: Judging Systems & UVA problem solving,	Discussion and Problem Solving	Problem Solving Ability and Written Examination	CO3, CO4
27-28 (Week 15)	ACM-ICPC: Judging Systems & UVA problem solving,			CO3, CO4
27-28 (Week 16)	Case Study & Review class for Semester Final Examination.	Lectures, Questions and answers	Problem Solving & Viva Voce	CO3, CO4

Final Exam (22 Nov -09 Dec 2024)

Rubrics						
COs (Bloom's Level)	Excellent (80%-100%)	Good (70%-79%)	Satisfactory (60%-69%)	Poor (40%-59%)	Unsatisfactory (0-39%)	Marks (70)
CO1 Understanding	Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.	Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.	Answer is brief with insufficient detail provided to support issues were introduced.	Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.	None of the relevant details were included or didn't answer.	10
CO2 Understanding	The question is answered appropriately explaining the problem solving strategy to a specific programming problem described in the question.	The question is answered briefly explaining the problem solving strategy to a specific programming problem described in the question.	The question is answered correctly by the problem solving strategy to a specific programming problem described in the question but some points are missing.	The question is answered incompletely explaining the problem solving strategy to a specific programming problem described in the question but some points are correct.	No attempt to explain the problem solving strategy to a specific programming problem described in the question method.	20
CO3 Analyzing	A clear, complete, and properly ordered chain of analyzing steps (i.e. perceive the problem scenario, relate the problem to the programming perception, explore the programming strategy to solve it) is followed to answer the question.	The chain of analyzing steps is complete and correctly ordered but lack of some expected plods.	One or more intermediate analyzing steps are missing or unclear, but the correctness of the analysis is not compromised.	One or more intermediate analyzing steps are missing or unclear to answer the question.	The stated chain of analysis does not lead to the stated question.	20

Teaching Strategy	Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. Multimedia projector and a PC will be used for the convenience of the students to understand codes practically. Students must participate in classroom discussions for case studies, problems solving and project developments.																																																		
Assessment and Marks Distribution	Continuous	Class Participation	10%	Grading Policy	A+ ≥ 80																																														
		Assignment / Presentation	10%		A 75 - < 80																																														
		Class Test	10%		A- 70 - < 75																																														
	Summative	Midterm Examination	30%		B+ 65 - < 70																																														
		Final Examination	40%		B 60 - < 65																																														
					B- 55 - < 60																																														
					C+ 50 - < 55																																														
					C 45 - < 50																																														
					D 40 - < 45																																														
					F < 40																																														
Additional Information	1. Academic Calendar: https://www.bubt.edu.bd/Home/page_details/Academic_Calendar 2. Rules & Regulations: https://www.bubt.edu.bd/Home/page_details/Rules_and_Regulations 3. Grading & Evaluation: https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System 4. Proctorial Rules: https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor																																																		
Overall CO Assessment Criteria	Assessment methods of COs are given below: <table> <tr> <th rowspan="2">Assessment Area</th><th colspan="4">CO</th><th rowspan="2">Assessment Area Mark</th></tr> <tr> <th>CO1</th><th>CO2</th><th>CO3</th><th>CO4</th></tr> <tr> <td>Class Participation</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr> <tr> <td>Assignment/Presentation</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr> <tr> <td>Class Test</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr> <tr> <td>Midterm Exam</td><td>10</td><td>10</td><td>10</td><td>-</td><td>30</td></tr> <tr> <td>Final Exam</td><td>-</td><td>10</td><td>10</td><td>20</td><td>40</td></tr> <tr> <td>Total Mark</td><td>10</td><td>20</td><td>20</td><td>20</td><td>70</td></tr> </table>					Assessment Area	CO				Assessment Area Mark	CO1	CO2	CO3	CO4	Class Participation	-	-	-	-	-	Assignment/Presentation	-	-	-	-	-	Class Test	-	-	-	-	-	Midterm Exam	10	10	10	-	30	Final Exam	-	10	10	20	40	Total Mark	10	20	20	20	70
Assessment Area	CO				Assessment Area Mark																																														
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Final Exam	-	10	10	20	40																																														
Total Mark	10	20	20	20	70																																														
Additional Course Policies	There will assignment. No late homework will be accepted. <u>Any kind of copy/manipulation in assignment will carry zero mark.</u> Two or more copied assignments will carry zero mark in all assignments. Zero tolerance will be shown in this regard. Solutions to assignment problems will be provided through web and on hand. There will be at least three class tests (CT). Best two of three or best three CTs will be counted. Both of regular and surprise CTs can be conducted. CT, Mid-term and final exam will be closed book, closed notes. Mobile phone is strictly prohibited in exam hall. Students are insisted to carry their own watch and synchronize time during exam hours. If a student is absent from class test anyway and made no report to the class teacher personally beforehand, his/her score for that test will be zero. No make-up for the class test will be allowed as 2 of 3 CTs are being considered. No make-up for Mid-exam will be entertained without physical presence and recommendation of the guardian along with written permission of the department. Make-up of Mid-exam may be much harder than the regular one.																																																		

CO4 Applying	The answer shows the complete and appropriate application of conceptual programming strategies to solve a problem.	The answer shows appropriate application of conceptual programming strategies incompletely to solve a problem.	The answer shows partially appropriate but complete application of conceptual programming strategies to solve a problem.	The answer shows partially appropriate application of conceptual programming strategies incompletely to solve a problem.	The answer shows the incomplete and inappropriate application of conceptual programming strategies to solve a problem. 20
Text Book	Teach Yourself C++ (Third Edition) by Herbert Schildt				
Reference Book	Supplementary Readings: A Complete Reference to C++ by Herbert Schildt				

Bloom's Taxonomy for Teaching-Learning

Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specificity. The three lists cover the learning objectives in Cognitive, Affective and Psychomotor domains. The Cognitive domain list has been the primary focus of most education and is frequently used to structure curriculum learning objectives, assessments and activities. The three domains and respective levels are illustrated below.



Descriptions of Cognitive Domain (Anderson and Krathwohl's Taxonomy 2001)

The **cognitive domain** involves the development of our mental skills and the acquisition of knowledge.

Level	Category	Meaning	Keywords
C1	Remembering	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.	Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, write
C2	Understanding	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, report
C3	Applying	Carrying out or using a procedure through executing, or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews or simulations.	Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, use
C4	Analyzing	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations.	Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, structure
C5	Evaluating	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Appraise, argue, assess, choose, conclude, critique, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, Monitor
C6	Creating	Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function.	Construct, design, develop, generate, hypothesize, invent, plan, produce, compose, create, make, perform, plan, produce

Graduate Attributes (Program Outcomes) for B.Sc. in Engineering Program based on Washington Accord

Program Outcomes (POs) are narrower statements that describe what students are expected to know and be able to do by the Time of graduation. These relate to the knowledge skills and attitudes that students acquire while progressing through the program. The students of the B.Sc. in CSE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation.

PO1–Engineering knowledge (Cognitive): Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2–Problem analysis (Cognitive): Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.

PO3–Design/development of solutions (Cognitive, Affective): Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.

PO4–Investigation (Cognitive, Psychomotor): Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

PO5–Modern tool usage (Psychomotor, Cognitive): Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6–The engineer and society (Affective): Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7–Environment and sustainability (Affective, Cognitive): Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8–Ethics (Affective): Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.

PO9–Individual work and teamwork (Psychomotor, Affective): Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.

PO10–Communication (Psychomotor, Affective): Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.

PO11–Project management and finance (Cognitive, Psychomotor): Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.

PO12–Life-long learning (Affective, Psychomotor): Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Prepared By:

Checked by:

Approved By:

